

Extended Block Working

Revision Guide

Module 4 | GERT8000-S5 (Iss 13, Dec 2025) | EMR PPT 36-05

1. What is Extended Block Working?

Extended Block Working (EBW) is a method of moving trains through a section where signalling has failed. It operates on the principle of one train in a section at a time.

Term	Meaning
EBW	Extended Block Working — replaces TBW and ESW (Dec 2025 rule book change)
NR3190	The EBW ticket form that records the details of each movement
TCB line	Track Circuit Block line — the type of line EBW applies to
Entrance signal	The signal at the start of the EBW section, kept at danger
Exit signal	The identifiable signal at the end of the EBW section
"End of degraded working" sign	Sign placed at the exit signal to mark the end of the EBW area

EBW is only authorised when:

- **Signalling equipment on a TCB line (not a single line) has failed or been disconnected, AND**
- **It is necessary to authorise drivers to pass two or more consecutive main aspect stop signals at danger under the same authority.**

2. The EBW Section

The EBW section runs between:

- The entrance signal — kept at danger on the approach to the affected area
- The exit signal — at a suitable, easily identifiable location beyond the affected area

Suitable exit signal locations include:

Station	Level crossing	Tunnel	Junction
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"End of Degraded Working" Sign

- Provided at the exit signal before EBW starts (wherever possible)
- EBW can start before the sign is provided — if the exit signal is at a station, level crossing, tunnel or junction
- The signaller will tell the driver whether the sign has been provided
- The sign may remain in place after EBW is withdrawn — a GSM-R message may advise the driver

3. At the Entrance Signal — The NR3190 Ticket

When your train is to enter the EBW section, the signaller will tell you where the exit signal is and whether the "end of degraded working" sign is in place. You must then complete an NR3190 ticket at the signaller's dictation.

Completing the Ticket

- Complete the NR3190 Extended Block Working Ticket at the signaller's dictation
- If more than one locomotive at the front: the driver in the leading cab completes the ticket, shows it to every other driver, and retains it during the movement

Before Entering

- Tell the guard (if provided) that EBW is in operation

You MUST operate the TPWS temporary isolation switch before entering the EBW section.

When No Ticket is Required

You will not receive or complete an NR3190 ticket if your train is entering the section to:

- Assist a failed train
- Evacuate passengers from a failed train
- Remove a portion of a divided train
- Remove a train or vehicles that have proceeded without authority

Important

- Even when no ticket is required, you MUST have authority from the signaller before entering the section
- If entering without a ticket, proceed at caution throughout the section

4. During the Movement

Follow the instructions shown on your NR3190 ticket throughout the movement.

Speed

Situation	Maximum speed
Normal EBW movement	50 mph (subject to any other restrictions)
First train — over points, switch diamonds or swing-nose crossings	15 mph
Entering without a ticket (assist, evacuate, etc.)	Proceed at caution

If You Are the First Train Through

- **Approach all points, switch diamonds and swing-nose crossings at caution**
- Check (if possible) that they are in the correct position
- Do not pass over any of them at more than 15 mph

The signaller will tell you if your train is the first through the section.

AWS at Signals in the EBW Section

AWS Rule during EBW
• AWS indications at signals you have been told to disregard must be acknowledged then ignored • You do not need to act on the signal aspect — but you must press the acknowledge button

5. Level Crossings within the EBW Section

- Details of any crossing attendants provided will be on the NR3190 ticket

A level crossing within the EBW section may only be passed over if the crossing attendant is displaying a green handsignal.

If no attendant is shown on the ticket, approach the crossing at caution and check it is safe before passing over it.

6. Arriving at the Exit Signal

When your train arrives at the exit signal, in this order:

1	STOP at the exit signal
2	Immediately reinstate TPWS
3	Tell the signaller your train has arrived at the exit signal
4	Write 'CANCELLED' across the NR3190 ticket
5	Obey the signaller's instructions for resuming normal working
6	Dispose of the cancelled ticket in accordance with company instructions

7. Key Numbers at a Glance

Rule / Limit	Value
Maximum speed in EBW section	50 mph
Maximum speed over points / crossings (first train)	15 mph
Lines where EBW applies	TCB lines (not single lines)
Minimum consecutive signals to justify EBW	2 or more
Form used to record EBW movements	NR3190
Level crossing — condition for passing	Crossing attendant shows green handsignal
Word written across ticket on completion	CANCELLED

Q & A — Extended Block Working

All questions sourced from GERT8000-S5 Issue 13 (December 2025) and EMR PPT 36-05.

Question
Q: What is Extended Block Working (EBW) and when is it authorised?
A: EBW is authorised when signalling equipment on a TCB line (not a single line) fails or is disconnected, and it is necessary for drivers to pass two or more consecutive main aspect stop signals at danger under the same authority. It is based on the principle of one train in a section.
Q: What is the NR3190 ticket?
A: The NR3190 is the Extended Block Working Ticket. It is completed by the driver at the signaller's dictation before entering the EBW section. It records the details of the movement including the exit signal location.
Q: If more than one locomotive is at the front of the train, who completes the NR3190 ticket?
A: The driver in the leading cab completes the ticket. They must show it to every other driver and then retain it during the movement.
Q: Before entering the EBW section, what two things must the driver do (apart from completing the ticket)?
A: 1. Tell the guard (if provided) that EBW is in operation. 2. Operate the TPWS temporary isolation switch.
Q: In which four circumstances is an NR3190 ticket NOT required to enter the EBW section?
A: 1. Assisting a failed train. 2. Evacuating passengers from a failed train. 3. Removing a portion of a divided train. 4. Removing a train or vehicles that have proceeded without authority. (In all these cases, you must still have authority from the signaller to enter.)
Q: What is the maximum speed in an EBW section?
A: 50 mph, subject to any other speed restrictions.
Q: If your train is the first through the EBW section, what must you do at points and crossings?
A: Approach all points, switch diamonds and swing-nose crossings at caution, check (if possible) that they are in the correct position, and not pass over any of them at more than 15 mph.
Q: What must you do with AWS at signals you have been told to disregard during EBW?
A: Acknowledge the AWS indication and then ignore it — do not act on the signal aspect.
Q: There is a level crossing within the EBW section. Under what condition may you pass over it?
A: Only if the crossing attendant is displaying a green handsignal. Details of any crossing attendants are shown on the NR3190 ticket.

Q: When your train arrives at the exit signal, list the actions in order.

A: 1. Stop at the exit signal. 2. Immediately reinstate TPWS. 3. Tell the signaller the train has arrived at the exit signal. 4. Write 'CANCELLED' across the NR3190 ticket. 5. Obey the signaller's instructions for resuming normal working. 6. Dispose of the ticket per company instructions.

Q: After EBW is withdrawn, what sign may still be in place and how might you be advised about it?

A: The "end of degraded working" sign may still be at the exit signal after EBW is withdrawn. You may be advised about it via a GSM-R message.

Q: What is the difference between the entrance signal and exit signal in an EBW section?

A: The entrance signal is kept at danger on the approach to the affected area. The exit signal is at a suitable identifiable location beyond the affected area (e.g. station, level crossing, tunnel or junction) and can be cleared for the train to proceed once the driver confirms arrival.

Q: Under what condition can EBW commence before the "end of degraded working" sign is provided at the exit signal?

A: When the exit signal is located immediately before, at, or immediately beyond a station, level crossing, tunnel or junction.